The standard CH employment-density index

Controls: Density measure—Ciccone–Hall index; CH specification—Standard CH; State (Ciccone–Hall).

1. Why use a county-based index? Two MSAs can have the same total employment per unit of land area even though employment is concentrated differently across their counties. The CH index retains this information. Let c denote a constituent county and g an MSA or state. County employment is n_c, land area is a_c, and employment density is $d_c = n_c/a_c$. Total regional employment is $N_g = \sum_{c \in g} n_c$. The index is

$$D_g(\theta) = \frac{1}{N_g} \sum_{c \in g} n_c^{\theta} a_c^{1-\theta} = \sum_{c \in g} \frac{n_c}{N_g} d_c^{\theta-1}.$$

Each county's density contribution is weighted by its employment share. The index depends on the unknown parameter θ, which the regression estimates.

2. The regression and reported coefficient. Write y_g for the selected log GDP per employment outcome, after any chosen outcome adjustments. With all industries and no adjustment, $y_g = \log(Q_g/N_g)$, where Q_g is total GDP. The model is

$$y_g = \alpha + \log D_g(\theta) + W_g'\delta + u_g.$$

Here W_g contains any selected regional controls and fixed effects, δ their coefficients, and u_g the error. The reported elasticity is $\rho = \theta - 1$, not θ. If all constituent county densities rose by 1% while their employment shares stayed fixed, the model predicts approximately ρ% higher GDP per employment. At $\rho = 0$, the standard index is one and its logarithm is zero. For one county, $\log D_g(\theta) = \rho \log d_c$, giving the ordinary county density equation.

3. How it is estimated. OLS selects θ by nonlinear least squares. IV uses the selected historical instrument in nonlinear GMM. A separate first-stage regression is unavailable for CH; the instrument-relevance statistic uses the fitted index's response to θ. IV interpretation still requires a relevant instrument and a valid exclusion restriction. If several nonlinear solutions exist, the app reports a warning and selects the solution nearest the nonlinear least-squares estimate.

4. Coverage and available settings. MSA CH permits regional controls, state fixed effects and separate outcome adjustments for schooling and college. Standard State CH uses all-industry GDP per employment, robust standard errors and no human-capital adjustment or state fixed effects. One observation per modern state leaves no variation after a separate effect for every state. Both models require complete constituent-county employment and land area; combined BEA county units remain together. Zero-employment counties contribute zero to the index.